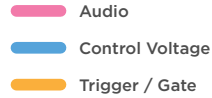
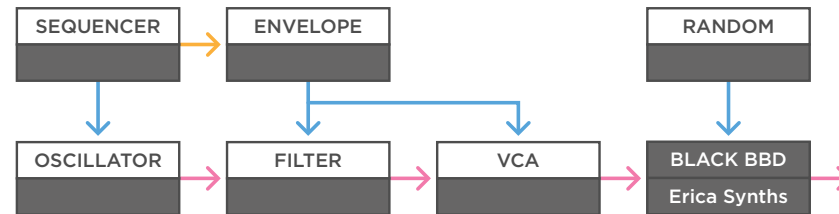


Erica Synths Black BBD

Classic Delay



MONOTRAIL TECH TALK



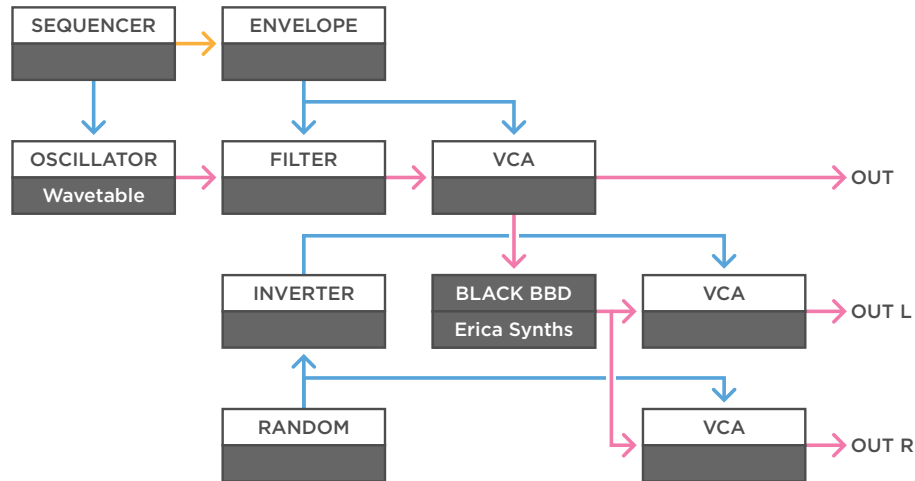
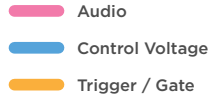
(a)

a - Here is a simple synth voice with an oscillator, filter, VCA, and a sequencer triggering an envelope to create a melody with short plucky sounds. Now you can use the Black BBD module as a traditional delay, by placing it at the end of the chain. You can set up the delay with a short delay time to create a slap back sound for example.

You can also experiment with longer delay times to create lingering delays, or very fast delay times for sound effects and reverb-like sounds. Try modulating different parameters of the delay, for example, a bit of random modulation to the mix CV input to create more dynamic effects.

Erica Synths Black BBD

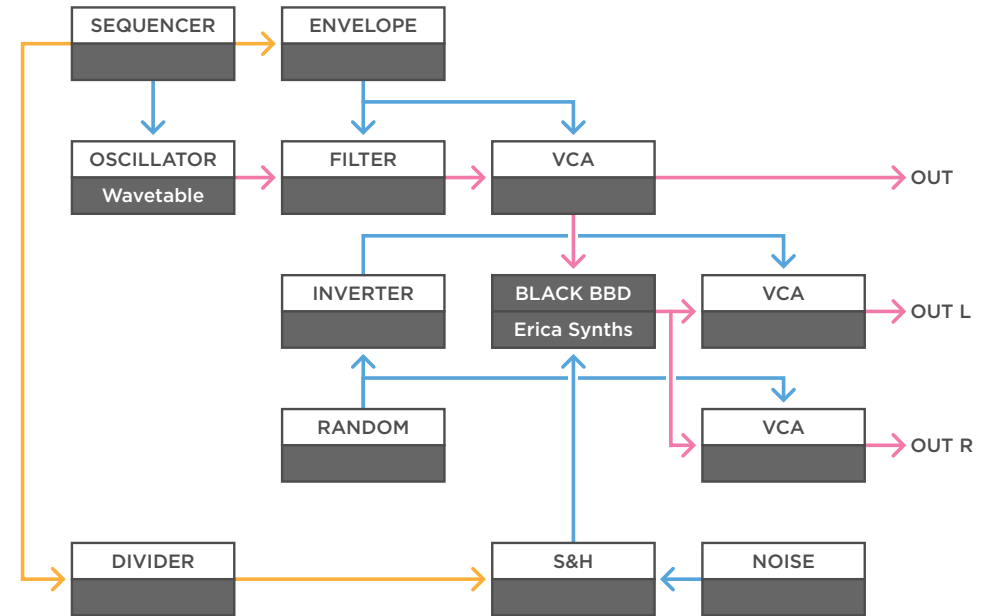
Classic Delay



(a)

a - This patch uses a wavetable oscillator through a filter and VCA. Again, with a sequencer to control the pitch of the oscillator, and an envelope to modulate the filter and VCA. But this time a slower melody is created. In order to create a nice spacious delay, you can send a copy of the dry signal from the VCA to the delay and multiply the wet output of the delay to two VCAs. Then use something like an LFO or random voltage to open and close those VCAs, in order to create a stereo effect. Don't forget to invert one of the signals.

MONOTRAIL TECH TALK



(b)

b - You can add extra modulation to this patch to create more movement. For example, use the gate output of the sequencer on a clock divider triggering a sample & hold module with some noise. Then you can use the stepped random signal to control the feedback. This works well with some more extreme settings.

Erica Synths Black BBD

Compositional Delay

- Audio
- Control Voltage
- Trigger / Gate

(a)
Sequence
Delay

C	E	G	C	C	E	G	C	C	E	G	C	C	E	G	C
	C	E	G	C	C	E	G	C	C	E	G	C	C	E	G
		C	E	G	C	E	G	C	C	E	G	C	C	E	G
			C	E	G	C	E	G	C	E	G	C	C	E	G

(b)
Sequence
Delay

C	F	E	A	E	B	A	E	C	F	E	A	E	B	A	E
		C	F	E	A	E	B	A	E	C	F	E	A	E	B
			F	A	E	B	A	E	B	A	E	B	A	E	B
				C	F	E	A	E	B	A	E	B	A	E	B

a - Beside using a delay to create depth to your sound, it's also a great tool for compositional use. For example, if you use a sequencer to play a few notes in a chord, and match the delay to the tempo, you can use it to build up chords.

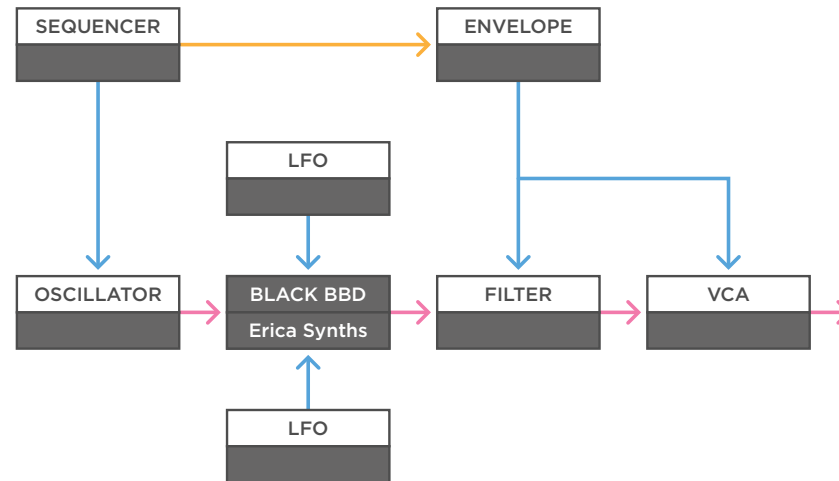
b - Or if you create a sequence and use a longer tempo matched delay time, you can create interesting melodies that interact with each other. You can have the delay repeat every 3rd note, for example, and use more feedback to create several layers.

Erica Synths Black BBD

Sound Shaping

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

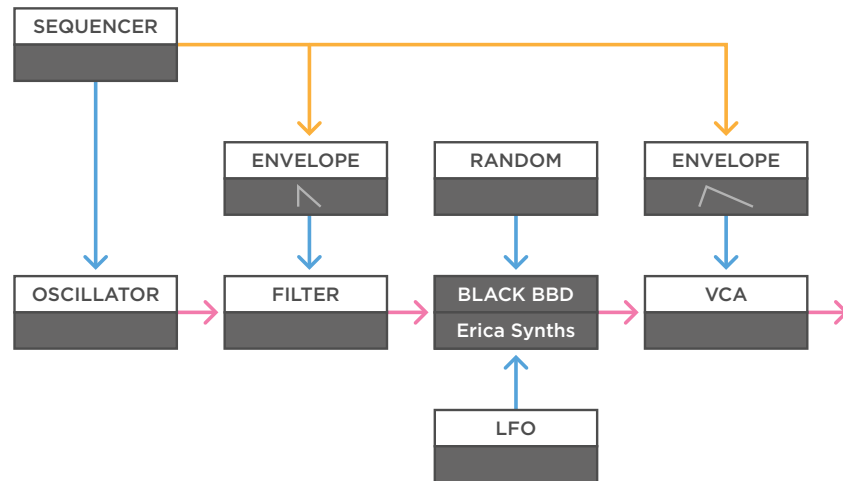
a - Instead of a classic approach, you can use a delay as a sound shaping device. For example, if you take an oscillator, you can place a delay right behind it to fatten up the single oscillator sound. This is a great way to turn a single saw wave into a multi-saw type sound. Just set the delay time pretty quick, and use something like a slow sine wave LFO to modulate the delay time slightly.

If you feed the result of this setup into a filter and VCA, and add a sequencer with an envelope, you have a lot of ways to create sonic variations. You can add to this patch by using an LFO to modulate the feedback input on the delay.

Erica Synths Black BBD

Sound Shaping

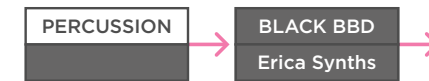
Audio
Control Voltage
Trigger / Gate



(a)

a - Experiment with a delay in different parts of your patch. If you set up a simple voice with oscillator and filter, and then run the sound through a delay before hitting the VCA, you can create interesting delay type effects that are still integrated in the sound volume of the voice. For example, use a sequencer to create a melody, and trigger both a short envelope that is used to shape the filter, and a longer envelope to control the volume on the VCA. Then add some random voltage to the delay time, and an LFO to modulate the feedback amount on the delay.

MONOTRAIL TECH TALK

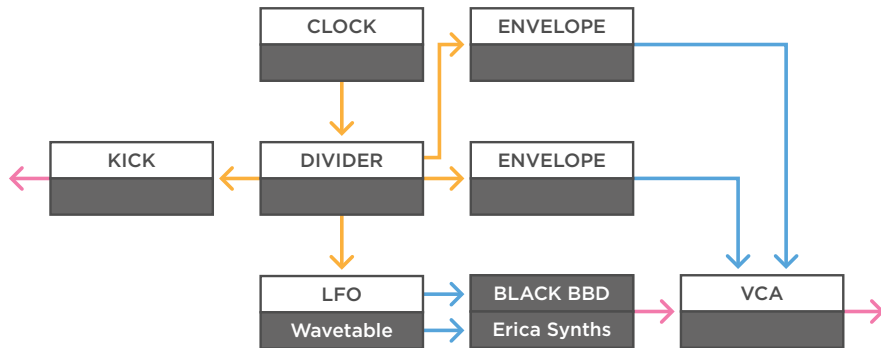
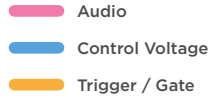


(b)

b - Short delay times can also work great on percussive sounds and drum loops. Just take something like a single hi-hat or sampled drum loop and run it through the delay. Tweaking the module by hand is also a great way to build tension and effects during live performances.

Erica Synths Black BBD

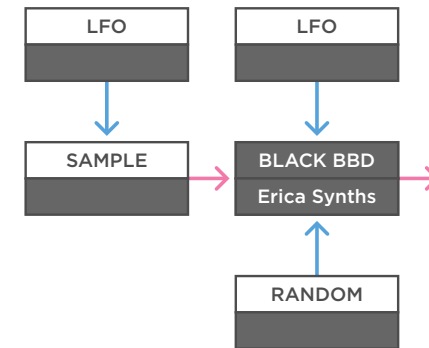
Textures & Drones



(a)

a - Delays that can self-oscillate can be used as sound sources. For example, you can use a delay as an industrial rhythmic sound source. Just crank up the feedback until it starts making sounds and add tempo synced modulation. Here you see a modular clock driving a clock divider. A fast division is used to trigger an envelope opening a VCA for a pulsating effect. A slower division is triggering a kick, as well as an inverted envelope, which is passed on to the same VCA for a side chain effect. Then another division is used to tempo sync a wavetable LFO, which is modulating the delay time and feedback amount on the delay module.

MONOTRAIL TECH TALK



(b)

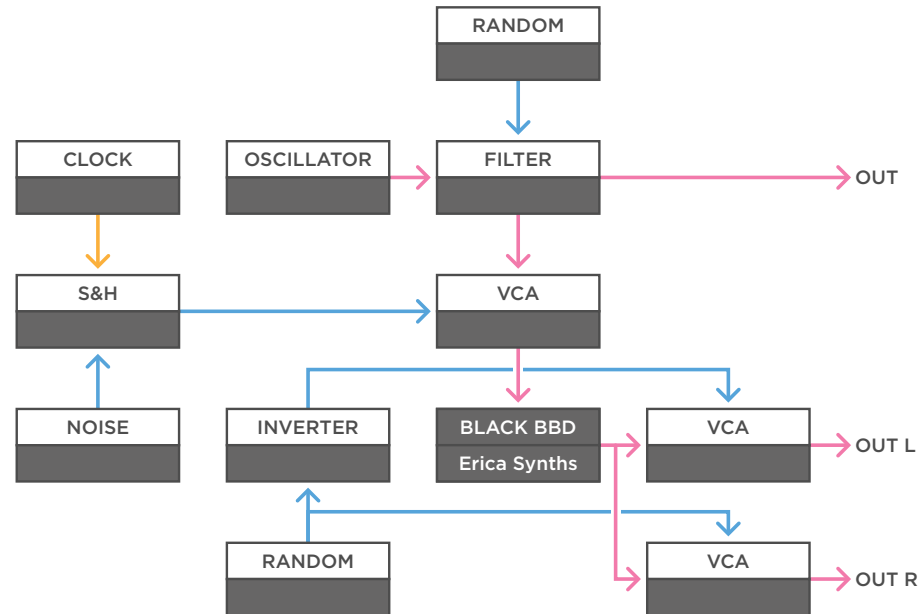
b - It's also worth it to experiment with running samples through the delay with higher feedback settings. Especially if the delay can go full wet, you can create other worldly effects by modulating the playback speed of the sample, as well as the delay time and feedback of the delay. Preferably all with independent modulation sources like random voltages or LFOs.

Erica Synths Black BBD

Textures & Drones

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

a - You can create interesting granular type effects if you feed a delay fragments of a sound, and set it to delay times that break up the sound nicely. For example, if you take a single oscillator, run it through a filter and modulate the filter with a slow random voltage, you create a simple constant drone. Make sure you send the audio to something like an outboard mixer so you can hear the dry sound.

Then multiply that audio signal and send it through a VCA. Add a sample & hold module and feed it some noise together with a steady clock to create a stepped random signal to open the VCA.

Because the noise is bipolar, and the VCA is not, a random 50% of the time the audio doesn't pass through the VCA, and the other 50% it will pass through, but with different volumes.

For some addition depth you can multiply the output of the delay to two VCAs for a stereo effect. Use another sample & hold circuit, or random voltage, to modulate the VCAs for a stereo spread. Then of course, add reverb to taste.